

PHYSICS EXAM — MARKING REPORT

SSC-I (9th Grade) — Annual Examination 2026 — Units 1 & 2

Assessment Date: 7 April 2026 | Report Generated: 7 April 2026

Student: Seemab	Subject: Physics	Paper: exam-01-ch1-2	Total Marks: 65
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TOTAL SCORE

63.5 / 65

97.7% — Grade A+

Section	Score	Maximum	Percentage
A — MCQs (12 questions, all attempted)	12	12	100%
B — Short Questions (11 attempted of 22)	32.5	33	98.5%
C — Long Questions (4 attempted of 8)	19	20	95%
TOTAL	63.5	65	97.7%

SECTION A — Multiple Choice Questions [12 × 1 = 12 Marks]

#	Question	Correct	Seemab	Mark
1	Which one of the following is not a derived unit?	B) kilogram	B	1
2	In SI, the base unit of temperature is:	B) kelvin	B	1
3	The SI derived unit of force (kg·m/s ²) is given the special name:	C) newton	C	1
4	The mass of Earth in scientific notation is:	A) 5.98×10^{24} kg	A	1
5	Which prefix represents the largest value?	C) peta	C	1
6	Which of the following is a vector quantity?	C) displacement	C	1
7	The least count of a vernier caliper (1 mm, 10 divisions):	B) 0.1 mm	B	1
8	The number of significant figures in 0.00580 km is:	B) 3	B	1
9	The motion of a rotating fan is an example of:	D) rotatory motion	D	1
10	A girl walks 3 km west and 4 km south. Distance and displacement:	D) 7 km, 5 km	D	1
11	In 5 s velocity increases by 20 m/s. Acceleration is:	B) 4 m/s ²	B	1
12	The slope of a distance-time graph gives the:	C) speed	C	1

Section A total: 12/12 (100%) — Perfect score.

SECTION B — Short Answer Questions [11 × 3 = 33 Marks] — All 11 counted

Q#	Topic	Score	Assessment
(i) OR	Non-physical quantities	3/3	Excellent Clear definition (cannot be measured by numerical value, subjective). Three valid examples: taste, feelings/emotions, beauty/colour perception. Good explanation of why they lack standard units.
(ii)	Seven SI base quantities & units	3/3	Excellent All seven listed correctly in a well-organized table: Length (l, m), Mass (m, kg), Time (t, s), Electric current (I, A), Temperature (T, K), Amount of substance (n, mol), Luminous intensity (I _v , cd).
(iii)	Scalar vs vector quantities	3/3	Excellent Clear definitions with difference in algebra rules (ordinary vs vector algebra). Gave 6 examples of each — more than the 3 required. All examples correct.
(iv)	Vernier caliper least count	3/3	Excellent Correct formula (LC = smallest main scale division / no. of vernier divisions). Calculation: $1/10 = 0.1 \text{ mm} = 0.01 \text{ cm}$. Good explanation of what least count means in practice.
(v)	Precision vs accuracy	3/3	Excellent Definitions well stated. Target/dart board diagrams effectively illustrate: (1) high precision + high accuracy, (2) high precision + low accuracy. Clear visual demonstration of the concept.
(vi)	Significant figures + scientific notation	3/3	Excellent All three parts correct: (a) 100.8 s = 4 SF, 1.008×10^2 s; (b) 0.00580 km = 3 SF, 5.80×10^3 km; (c) 210.0 g = 4 SF, 2.10×10^2 g.
(vii)	Distance & displacement (circular track)	3/3	Excellent Clear definitions (distance = total path, scalar; displacement = shortest path, vector). Runner: distance = 400 m, displacement = 0 m with good reasoning (returns to start).
(viii) OR	Average vs instantaneous speed	3/3	Excellent Good definitions and practical comparison (speedometer analogy). Calculation: $70 \text{ km} \div 1 \text{ hr} = 70 \text{ km/h} = 19.44 \text{ m/s}$. Conversion shown step by step.
(ix) OR	Maglev train speed	3/3	Excellent Data extraction correct (30 km, 450 s). Speed in m/s = 66.67 m/s. Speed in km/h = 240 km/h. Both conversions shown clearly.
(x) OR	Cheetah acceleration	2.5/3	Note Data and final answer correct (8.93 m/s ²). However, formula was written as “Initial velocity – Final velocity / time” which is backwards . Self-corrected in working but the stated formula is wrong. Always write a

			$= (v_f - v_i) / t.$
(xi) OR	Distance-time graphs	3/3	Excellent All three graphs correctly drawn: (a) body at rest = horizontal line, (b) constant speed = straight diagonal, (c) increasing speed = upward curve. All axes properly labeled.

Section B total: 3 + 3 + 3 + 3 + 3 + 3 + 3 + 3 + 3 + 3 + 2.5 + 3 = 32.5/33 (98.5%) — Outstanding performance across all 11 questions.

SECTION C — Long Questions [4 × 5 = 20 Marks] — All 4 counted

Q#	Topic	Score	Assessment
Q3	Vernier caliper: diagram, method, zero error, LC	4/5	Good (a) Definition and labeled diagram with all 6 parts (outside jaws, inside jaws, main scale, vernier scale, retainer, depth probe). (b) Measurement method described step-by-step, zero error types (none, positive, negative) well explained. (c) Formula for LC given correctly, but did not calculate for 20 divisions as asked (answer should be $1/20 = 0.05$ mm). Lost 1 mark.
Q4 OR	Scientific notation, conversions, prefixes	5/5	Excellent (a) Definition (mantissa × 10^n) and 4 clear reasons for use in physics. (b) All 3 conversions correct: 1.4953×10^{11} m = 149.53 Gm; 8.0×10^{-4} g = 0.8 mg; 8.64×10^4 s = 86.4 ks. (c) Prefixes definition + table of 6 prefixes (Giga, Mega, Kilo, Milli, Micro, Nano) with symbols, multipliers, and examples. Comprehensive answer.
Q5 OR	Speed-time graph, acceleration, total distance	5/5	Excellent (a) Graph correctly drawn (trapezoid shape) with labeled axes and phases. (b) All 3 accelerations correct: A→B = 1.5 m/s^2 , B→C = 0 m/s^2 , C→D = -3 m/s^2 . (c) Area under graph calculated by breaking into shapes: $300 + 900 + 150 = 1350$ m. Average speed = 22.5 m/s . Perfect working.
Q6	Speed/velocity/acceleration + Shoaib Akhtar + falling block	5/5	Excellent (a) All 3 defined with formulae and vector notation (speed = scalar, velocity & acceleration = vectors). (b) Conversion $161.3 \text{ km/h} = 44.81 \text{ m/s}$, reaction time = 0.39 s . (c) Block: $t = (78.5 - 0)/9.8 = 8.01 \text{ s}$. All parts flawless.

Section C total: 4 + 5 + 5 + 5 = 19/20 (95%) — Excellent across all 4 questions.

Overall Assessment

Score: 63.5/65 (97.7%) — Near-perfect result. Excellent command of both Unit 1 (Physical Quantities & Measurement) and Unit 2 (Kinematics). Lost only 1.5 marks across the entire paper.

Strengths:

- Perfect MCQ score — solid conceptual understanding
- Numerical problems solved accurately with clear step-by-step working (Maglev, Shoaib Akhtar, falling block, speed-time graph)
- Effective use of diagrams (target boards, distance-time graphs, speed-time graph, vernier caliper)
- Strong across all sections with consistent quality

Marks lost (1.5 total):

- **Q2(x) OR — Formula direction (−0.5):** Wrote acceleration formula as $(v_i - v_f)/t$ instead of $(v_f - v_i)/t$. Self-corrected in working, but the stated formula is wrong. Practice writing $a = (v_f - v_i) / \Delta t$ consistently.
- **Q3(c) — Incomplete calculation (−1):** Formula for least count given correctly, but did not substitute values for 20 divisions (should be $1/20 = 0.05$ mm). Always complete the numerical calculation when a specific answer is asked.